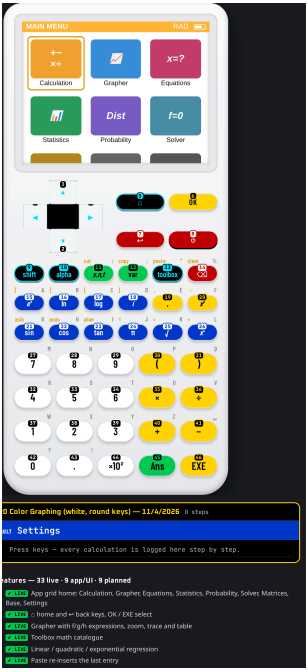
U-45 CALCULATORS

User Manual

Calculator #20 Color Graphing (white, round keys)

Graphing – colour



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#20 — Color Graphing (white, round keys)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example [#1 ▲]. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

algebraic with history (type the expression, then EXE / ENTER).

> This model starts on its **main menu**. Press [#37] [1] to open the calculation app first; the examples include this step.

Chapters

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5. Worked examples
6. Tricks & hacks
7. Keyboard shortcuts
8. Common mistakes
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1 · Getting started

Open U-45 CALCULATORS, choose **#20** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#27, #28, #29, #32, #33, #34, #37, #38, #39, #42, #43, #44
op	Operators & execute	#6, #19, #20, #30, #31, #35, #36, #40, #41, #46
fn	Functions	#15, #16, #17, #18, #21, #22, #23, #24, #25, #26
mem	Memory, variables & registers	#11, #12, #45
clr	Clear, delete & power	#7, #8, #14
mod	Shift, modes, menus & navigation	#1, #2, #3, #4, #5, #9, #10, #13

2 · Features

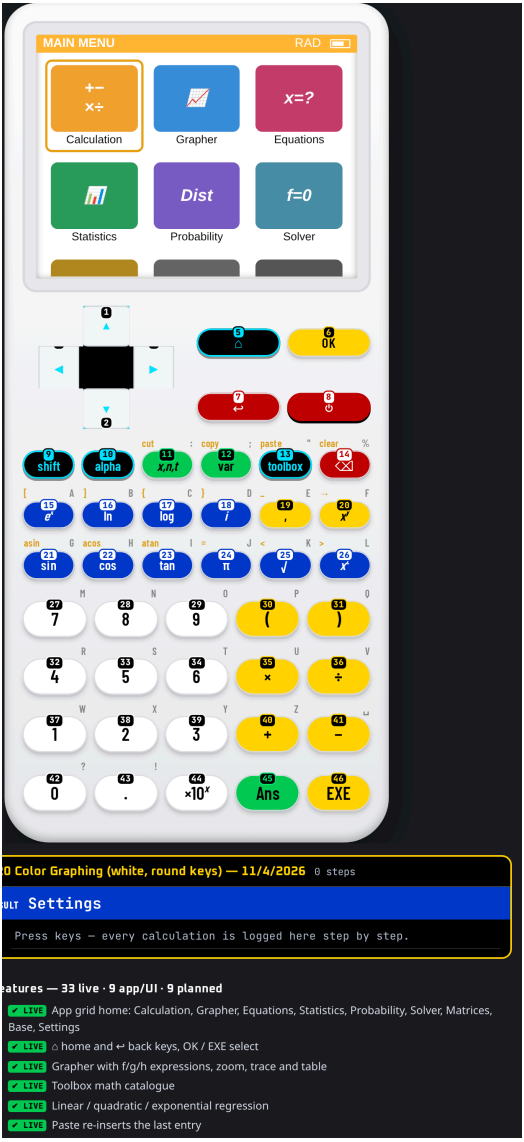
51 features: 33 live on the keys, 9 app/UI features, 9 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE App grid home: Calculation, Grapher, Equations, Statistics, Probability, Solver, Matrices, Base, Settings
2. LIVE ⌂ home and ↩ back keys, OK / EXE select
3. LIVE Grapher with f/g/h expressions, zoom, trace and table
4. LIVE Toolbox math catalogue
5. LIVE Linear / quadratic / exponential regression
6. LIVE Paste re-inserts the last entry
7. LIVE Variables A–Z via alpha
8. LIVE Square root $\sqrt{}$
9. LIVE Square x^2
10. LIVE Power x^y / ^ with right-to-left chaining (2^3^2 = 512)
11. LIVE Common logarithm $\log_{10}$
12. LIVE Natural logarithm ln
13. LIVE Exponential e^x
14. LIVE Trigonometry sin / cos / tan honouring the current angle unit
15. LIVE Inverse trigonometry $\sin^{-1}$ / $\cos^{-1}$ / $\tan^{-1}$ with domain checking (Math ERROR outside −1…1)
16. LIVE Factorial x! (exact up to 69!)
17. LIVE π constant to full internal precision
18. LIVE Scientific-notation entry ×10^x / EXP / EEX

19. LIVE Ans recall of the last result; operators auto-insert Ans
20. LIVE Store a value into memory / a variable (STO, sto►, →, Min)
21. LIVE Named variables (A–F, X, Y, M … or A–Z) usable inside expressions
22. LIVE SOLVE — numeric equation solver (Newton / bisection)
23. LIVE TABLE — tabulate f(x) over start / end / step
24. LIVE Statistics: data entry, n, $\bar{x}$, σ, s, Σx, Σx^2
25. LIVE Two-variable statistics and linear regression (a, b, r)
26. LIVE Matrix mode: define, add, multiply, det, transpose, identity
27. LIVE Complex numbers: i, arg, conjugate, Re / Im, polar ↔ rectangular
28. LIVE BASE-N: DEC / HEX / BIN / OCT conversions
29. LIVE Bitwise logic AND / OR / XOR / XNOR / NOT
30. LIVE Backspace / DEL editing of the current entry
31. LIVE Replay ▲ ▼ — scroll back to previous calculations and re-edit them
32. LIVE MATH / Toolbox menu of templates and functions
33. LIVE Probability distributions (normal, binomial, Poisson) and PRB menu
34. APP/UI Every key numbered 1…N in reading order — matches the key map in the user manual
35. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a “Classic” toggle
36. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)
37. APP/UI Show / hide key numbers with one button (remembered between sessions)
38. APP/UI Copy the current display text to the clipboard

39. APP/UI **Physical-keyboard shortcuts: digits, + – * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i**
40. APP/UI **Fit / 1:1 zoom that never pushes keys off a phone screen**
41. APP/UI **Reference photo viewer and tool-tip name on every key**
42. APP/UI **Reset = fresh power-on state; works fully offline (PWA cache / Android APK)**
43. PLANNED **Python scripts**
44. PLANNED **Sequences app**
45. PLANNED **More regression models**
46. PLANNED **Cut / copy clipboard**
47. PLANNED **Graph image export (PNG)**
48. PLANNED **Shareable function list link**
49. PLANNED **Dark screen theme**
50. PLANNED **Speech read-out of values**
51. PLANNED **Exam mode**

3 · Numbered key map



Calculator #20 with key numbers shown (Hi-Contrast colours).

Key table

“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	▲	·	·	·	mod
2	▼	·	·	·	mod
3	◀	·	·	·	mod
4	▶	·	·	·	mod
5	⌂	·	·	·	mod
6	OK	·	·	·	op
7	↩	·	·	·	clr
8	⏻	·	·	·	clr
9	shift	·	·	·	mod
10	alpha	·	·	·	mod
11	x,n,t	cut	:	·	mem
12	var	copy	;	·	mem
13	toolbox	paste	"	·	mod
14	⌫	clear	%	·	clr
15	ex	[	A	·	fn
16	ln	]	B	·	fn
17	log	{	C	·	fn
18	i	}	D	·	fn
19	,	_	E	·	op
20	xy	→	F	·	op
21	sin	asin	G	·	fn
22	cos	acos	H	·	fn
23	tan	atan	I	·	fn

#	Face	Shift	Alpha	3rd	Group
24	π	=	J	·	fn
25	√	<	K	·	fn
26	x^2	>	L	·	fn
27	7	·	M	·	num
28	8	·	N	·	num
29	9	·	O	·	num
30	(	·	P	·	op
31	)	·	Q	·	op
32	4	·	R	·	num
33	5	·	S	·	num
34	6	·	T	·	num
35	×	·	U	·	op
36	÷	·	V	·	op
37	1	·	W	·	num
38	2	·	X	·	num
39	3	·	Y	·	num
40	+	·	Z	·	op
41	−	·	␣	·	op
42	0	·	?	·	num
43	.	·	!	·	num
44	×10x	·	·	·	num
45	Ans	·	·	·	mem
46	EXE	·	·	·	op

4 · Key-by-key reference

Every key on calculator #20, in number order, with every function it carries in this app.

#1 ▲ `Shift, modes, menus & navigation`

Primary: Cursor ▲ — up / previous history line / previous item

#2 ▼ `Shift, modes, menus & navigation`

Primary: Cursor ▼ — down / next history line / next item

#3 ◀ `Shift, modes, menus & navigation`

Primary: Cursor ◀ — move left in the line

#4 ▶ `Shift, modes, menus & navigation`

Primary: Cursor ▶ — move right in the line

#5 ⌂ `Shift, modes, menus & navigation`

Primary: HOME — return to the main calculation screen

#6 OK `Operators & execute`

Primary: Execute / = — evaluate the line and show the result (EXE / ENTER / =)

Keyboard: `Enter` `=`

#7 ↩ `Clear, delete & power`

Primary: QUIT — leave the current screen / app

#8 ⏻ `Clear, delete & power`

Primary: OFF — power off (state is lost; press ON to restart)

#9 **shift** `Shift, modes, menus & navigation`

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#10 **alpha** `Shift, modes, menus & navigation`

Primary: ALPHA — the next key types its letter/variable (red/green label)

#11 **x,n,t** `Memory, variables & registers`

Primary: The graphing variable (X / x,θ,t,n) for Y= functions, tables and solving

alpha layer (press #10 first): : — Colon : (multi-statement separator)

#12 **var** `Memory, variables & registers`

Primary: VARS — variables menu

#13 toolbox `Shift, modes, menus & navigation`

Primary: MATH menu — extra math functions

shifted (2nd / f / SHIFT layer) (press #9 first): paste — ENTRY — recall a previous entry

#14 ⌫ `Clear, delete & power`

Primary: DEL — delete the character at the cursor

shifted (2nd / f / SHIFT layer) (press #9 first): clear — CLEAR — clear the entry line / screen

Keyboard: `Backspace` `Delete`

#15 ex `Functions`

Primary: e^x — exponential

shifted (2nd / f / SHIFT layer) (press #9 first): [— Types "["

alpha layer (press #10 first): A — Variable A (store with STO/ →, recall by name)

#16 ln `Functions`

Primary: Natural logarithm ln

shifted (2nd / f / SHIFT layer) (press #9 first):] — Types "]"

alpha layer (press #10 first): B — Variable B (store with STO/ →, recall by name)

Keyboard: `n`

#17 log Functions

Primary: Common logarithm $\log_{10}$

shifted (2nd / f / SHIFT layer) (press #9 first): { — Types "{"

alpha layer (press #10 first): C — Variable C (store with STO/ →, recall by name)

Keyboard: [l]

#18 i Functions

Primary: Imaginary unit i (complex numbers)

shifted (2nd / f / SHIFT layer) (press #9 first): } — Types "}"

alpha layer (press #10 first): D — Variable D (store with STO/ →, recall by name)

#19 , Operators & execute

Primary: Comma / argument separator (between function arguments, list items or coordinates)

alpha layer (press #10 first): E — Variable E (store with STO/ →, recall by name)

Keyboard: [,]

#20 xy Operators & execute

Primary: Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

shifted (2nd / f / SHIFT layer) (press #9 first): → — STO▶ (→) — store into a variable

alpha layer (press #10 first): F — Variable F (store with STO/ →, recall by name)

Keyboard: [^]

#21 sin Functions

Primary: Sine (uses the current angle unit)

shifted (2nd / f / SHIFT layer) (press #9 first): asin — Inverse sine $\sin^{-1}$

alpha layer (press #10 first): G — Variable G (store with STO/ →, recall by name)

Keyboard: s

#22 cos Functions

Primary: Cosine (current angle unit)

shifted (2nd / f / SHIFT layer) (press #9 first): acos — Inverse cosine $\cos^{-1}$

alpha layer (press #10 first): H — Variable H (store with STO/ →, recall by name)

Keyboard: c

#23 tan Functions

Primary: Tangent (current angle unit)

shifted (2nd / f / SHIFT layer) (press #9 first): atan — Inverse tangent $\tan^{-1}$

alpha layer (press #10 first): I — Variable I (store with STO/ →, recall by name)

Keyboard: t

#24 π Functions

Primary: π (3.14159265…) as an exact constant

shifted (2nd / f / SHIFT layer) (press #9 first): = — Equals — complete the calculation

alpha layer (press #10 first): J — Variable J (store with STO/ →, recall by name)

Keyboard: p

#25 √ Functions

Primary: Square root √

alpha layer (press #10 first): K — Variable K (store with STO/ →, recall by name)

Keyboard: r

#26 x^2 Functions

Primary: Square x^2

alpha layer (press #10 first): L — Variable L (store with STO/ →, recall by name)

Keyboard: q

#27 7 Digits & number entry

Primary: Digit 7

alpha layer (press #10 first): M — Variable M (store with STO/ →, recall by name)

Keyboard: 7

#28 8 Digits & number entry

Primary: Digit 8

alpha layer (press #10 first): N — Variable N (store with STO/ →, recall by name)

Keyboard: 8

#29 **9** Digits & number entry

Primary: Digit 9

alpha layer (press #10 first): 0 — Variable 0 (store with STO/ →, recall by name)

Keyboard: 9

#30 **(** Operators & execute

Primary: Opening bracket (

alpha layer (press #10 first): P — Variable P (store with STO/ →, recall by name)

Keyboard: (

#31 **)** Operators & execute

Primary: Closing bracket). Closing brackets at the end of a line may be omitted

alpha layer (press #10 first): Q — Variable Q (store with STO/ →, recall by name)

Keyboard:)

#32 **4** Digits & number entry

Primary: Digit 4

alpha layer (press #10 first): R — Variable R (store with STO/ →, recall by name)

Keyboard: 4

#33 **5** Digits & number entry

Primary: Digit 5

alpha layer (press #10 first): S — Variable S (store with STO/ →, recall by name)

Keyboard: 5

#34 **6** Digits & number entry

Primary: Digit 6

alpha layer (press #10 first): T — Variable T (store with STO/ →, recall by name)

Keyboard: 6

#35 **×** Operators & execute

Primary: Multiplication ×

alpha layer (press #10 first): U — Variable U (store with STO/ →, recall by name)

Keyboard: * x

#36 **÷** Operators & execute

Primary: Division ÷

alpha layer (press #10 first): V — Variable V (store with STO/ →, recall by name)

Keyboard: /

#37 **1** `Digits & number entry`

Primary: Digit 1

alpha layer (press #10 first): W — Variable W (store with STO/ →, recall by name)

Keyboard: 1

#38 **2** `Digits & number entry`

Primary: Digit 2

alpha layer (press #10 first): X — Variable X (store with STO/ →, recall by name)

Keyboard: 2

#39 **3** `Digits & number entry`

Primary: Digit 3

alpha layer (press #10 first): Y — Variable Y (store with STO/ →, recall by name)

Keyboard: 3

#40 **+** `Operators & execute`

Primary: Addition +

alpha layer (press #10 first): Z — Variable Z (store with STO/ →, recall by name)

Keyboard: +

#41 **−** `Operators & execute`

Primary: Subtraction −

Keyboard: -

#42 0 Digits & number entry

Primary: Digit 0

Keyboard: 0

#43 . Digits & number entry

Primary: Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

alpha layer (press #10 first): ! — Factorial n! (also on non-integers via the gamma function where supported)

Keyboard: .

#44 ×10x Digits & number entry

Primary: ×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23

Keyboard: e

#45 Ans Memory, variables & registers

Primary: Ans — recall the last answer into the current expression

#46 EXE Operators & execute

Primary: Execute / = — evaluate the line and show the result (EXE / ENTER / =)

5 · Worked examples

17 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Order of operations

Calculate 12 + 3 × 4. Press Reset first.

#37 1 *(open the calculation app from the main menu)* → #37 1 → #38 2 → #40 + → #39 3 → #35 × → #32 4 → #6 OK

Display shows 24 **— expected 24. ✔ verified on the app.**

Multiplication is done before addition, so the calculator holds the + until the × part is finished.

Example 5.2 — Brackets first

Calculate (12 + 8) ÷ 4. Press Reset first.

#37 1 *(open the calculation app from the main menu)* → #30 (→ #37 1 → #38 2 → #40 + → #28 8 → #31) → #36 ÷ → #32 4 → #6 OK

Display shows 5 **— expected 5. ✔ verified on the app.**

Brackets force the sum to be worked out before the division.

Example 5.3 — Two brackets multiplied

Calculate (3 + 4) × (9 − 2). Press Reset first.

#37 1 *(open the calculation app from the main menu)* → #30 (→ #39 3 → #40 + → #32 4 → #31) → #35 × → #30 (→ #29 9 → #41 − → #38 2 → #31) → #6 OK

Display shows 49 **— expected 49. ✔ verified on the app.**

Both brackets are evaluated first, then multiplied.

Example 5.4 — Squares

Calculate 12^2. Press Reset first.

#37 1 *(open the calculation app from the main menu)* → #37 1 → #38 2 → #26 x^2 → #6 OK

Display shows 144 **— expected 144. ✔ verified on the app.**

x^2 squares the value — quicker and safer than typing the number twice.

Example 5.5 — Square roots

Calculate √144. Press Reset first.

#37 1 *(open the calculation app from the main menu)* → #25 √ → #37 1 → #32 4 → #32 4 → #6 OK

Display shows 12 **— expected 12. ✔ verified on the app.**

The square-root key returns the principal (positive) root.

Example 5.6 — Powers

Calculate 2^10. Press Reset first.

#37 1 *(open the calculation app from the main menu)* → #38 2 → #20 x^y → #37 1 → #42 0 → #6 OK

Display shows 1024 **— expected 1024. ✔ verified on the app.**

The power key raises the first number to the second.

Example 5.7 — Trigonometry (degrees)

Calculate cos 0° (Deg mode). Press Reset first.

#37 1 *(open the calculation app from the main menu)* → #22 cos → #42 0 → #31) → #6 OK

Display shows 1 **— expected 1. ✔ verified on the app.**

Trig functions use the current angle unit; these answers assume degrees (the start-up setting).

Example 5.8 — Common logarithm

Calculate log 1000. Press Reset first.

#37 1 *(open the calculation app from the main menu)* → #17 log → #37 1 → #42 0 → #42 0 → #42 0 → #31) → #6 OK

Display shows 3 **— expected 3. ✔ verified on the app.**

log is base-10: it counts the zeros of a power of ten.

Example 5.9 — Natural logarithm

Calculate **ln 1**. Press **Reset** first.

#37 1 *(open the calculation app from the main menu)* → #16 ln → #37 1 → #31) → #6 OK

Display shows `0` — expected 0. ✔ verified on the app.

ln is the natural (base-e) logarithm.

Example 5.10 — Scientific notation

Calculate **6×10^2 × 3**. Press **Reset** first.

#37 1 *(open the calculation app from the main menu)* → #34 6 → #44 ×10^x → #38 2 → #35 × → #39 3 → #6 OK

Display shows `1800` — expected 1800. ✔ verified on the app.

The exponent key enters ×10^n without typing × 10 ^.

Example 5.11 — Area of a circle πr^2

Calculate **π × 2.5^2**. Press **Reset** first.

#37 1 *(open the calculation app from the main menu)* → #24 π → #35 × → #38 2 → #43 . → #33 5 → #26 x^2 → #6 OK

Display shows `19.63495408` — expected 19.63495408. ✔ verified on the app.

π is stored to full precision — never type 3.14.

Example 5.12 — Factorials

Calculate 5!. Press Reset first.

#37 1 *(open the calculation app from the main menu)* → **#33** 5 → **#10** alpha → **#43** ! → **#6** OK

Display shows `120` **— expected 120. ✔ verified on the app.**

n! multiplies 1×2×…×n; 0! is 1 by definition.

Example 5.13 — Continue from the answer

Calculate 100 × 4 = , then Ans ÷ 2. Press Reset first.

#37 1 *(open the calculation app from the main menu)* → **#37** 1 → **#42** 0 → **#42** 0 → **#35** × → **#32** 4 → **#6** OK → **#36** ÷ → **#38** 2 → **#6** OK

Display shows `200` **— expected 200. ✔ verified on the app.**

Pressing an operator straight after a result continues from that answer (Ans).

Example 5.14 — Average of three numbers

Calculate **(4 + 8 + 15) ÷ 3**. Press **Reset** first.

#37 1 *(open the calculation app from the main menu)* → #30 (→ #32 4 → #40 + → #28 8 → #40 + → #37 1 → #33 5 → #31) → #36 ÷ → #39 3 → #6 OK

Display shows 9 — expected 9. ✔ verified on the app.

Add up the values, then divide by how many there are.

Example 5.15 — Mixed operations

Calculate **50 − 4 × 6 + 7**. Press **Reset** first.

#37 1 *(open the calculation app from the main menu)* → #33 5 → #42 0 → #41 − → #32 4 → #35 × → #34 6 → #40 + → #27 7 → #6 OK

Display shows 33 — expected 33. ✔ verified on the app.

Only the × part is done first; the − and + are then worked left to right.

Example 5.16 — Dividing twice

Calculate **360 ÷ 4 ÷ 9**. Press **Reset** first.

#37 1 *(open the calculation app from the main menu)* → #39 3 → #34 6 → #42 0 → #36 ÷ → #32 4 → #36 ÷ → #29 9 → #6 OK

Display shows `10` — expected 10. ✔ verified on the app.

Division is worked left to right: the first quotient is divided again.

Example 5.17 — Brackets inside a longer formula

Calculate **((3 + 5) × 4 − 2) ÷ 10**. Press **Reset** first.

#37 1 *(open the calculation app from the main menu)* → #30 (→ #30 (→ #39 3 → #40 + → #33 5 → #31) → #35 × → #32 4 → #41 − → #38 2 → #31) → #36 ÷ → #37 1 → #42 0 → #6 OK

Display shows `3` — expected 3. ✔ verified on the app.

The innermost brackets are worked out first, then the outer brackets, then the division.

6 · Tricks & hacks

Hack 1: History reuse

Scroll up in the home screen history and press ENTER/EXE to copy an old entry or answer.

Hack 2: Ans chaining

Start an entry with an operator to continue from the previous answer.

Hack 3: Store variables

Use STO▶ / → to store values into letters A–Z, then use them in expressions.

Hack 4: Table of values

Use the TABLE view to scan where a function changes sign before solving.

Hack 5: Window first

If a graph looks empty, adjust the WINDOW / zoom standard.

Hack 6: Edit instead of retyping

After a result, press ◀ or ▶ to bring the last expression back with the cursor in it; change one number and press = again.

Hack 7: Ans chains

Start a line with an operator (e.g. ×) and the calculator inserts Ans automatically — great for compound growth: 1.05 ×, then = = = ...

Hack 8: Exact vs decimal

Natural-display models keep fractions, √ and π exact. Press S⇔D to flip to decimal; press it again to return.

Hack 9: Close brackets for free

Trailing closing brackets can be left off; = closes them for you.

Hack 10: Implicit multiplication

Type 2π or 3(4+1) without ×. Implicit multiplication binds tighter than ÷ on textbook models — use brackets if in doubt.

Hack 11: Store constants

Store a value you reuse (e.g. g = 9.81) with STO and a letter; then type the letter in formulas.

Hack 12: Power chains

x^y chains right to left: 2^3^2 = 2^9 = 512, not 64. Use brackets: (2^3)^2 = 64.

Hack 13: Angles

Check the D / R / G indicator before trig. sin 30 = 0.5 only in degrees.

Hack 14: Memory sums

M+ / M− keep a running total separate from the calculation line; RCL M shows it.

Hack 15: Replay history

Use ▲ ▼ to scroll earlier lines; press = on an old line to re-run it.

Hack 16: Hidden functions on this model

39 extra functions sit on shifted layers. Press the layer key once, then the key: for example #10 then #11 → :; #9 then #13 → paste; #9 then #14 → clear; #9 then #15 → [.

Hack 17: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left→right, top→bottom from #1 to #46.

Hack 18: Type faster on a computer

30 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 19: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 20: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 21: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
`Enter` `=`	#6	OK	Execute / = — evaluate the line and show the result (EXE / ENTER / =)
`Backspace` `Delete`	#14	⌫	DEL — delete the character at the cursor
`n`	#16	ln	Natural logarithm ln
`l`	#17	log	Common logarithm $\log_{10}$
`,`	#19	,	Comma / argument separator (between function arguments, list items or coordinates)
`^`	#20	xy	Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)
`s`	#21	sin	Sine (uses the current angle unit)
`c`	#22	cos	Cosine (current angle unit)
`t`	#23	tan	Tangent (current angle unit)
`p`	#24	π	π (3.14159265…) as an exact constant
`r`	#25	√	Square root √
`q`	#26	x^2	Square x^2
`7`	#27	7	Digit 7
`8`	#28	8	Digit 8
`9`	#29	9	Digit 9
`(`	#30	(	Opening bracket (
`)`	#31	)	Closing bracket). Closing brackets at the end of a line may be omitted
`4`	#32	4	Digit 4
`5`	#33	5	Digit 5

6	#34	6	Digit 6
* x	#35	×	Multiplication ×
/	#36	÷	Division ÷
1	#37	1	Digit 1
2	#38	2	Digit 2
3	#39	3	Digit 3
+	#40	+	Addition +
-	#41	−	Subtraction −
0	#42	0	Digit 0
.	#43	.	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored
e	#44	×10x	×10^{x} / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10^{x} 23

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: (–) vs –

Use the negative key for negative numbers.

Mistake 2: Radian mode

Graphs of sin x in degrees look flat over −10..10; switch to radians for standard trig windows.

Mistake 3: Implicit multiplication

2/3x may be read as 2/(3x) depending on the model; use brackets.

Mistake 4: Using – instead of (–)

The negative key (–) and the subtraction key are different; "– 5" at the start uses Ans – 5.

Mistake 5: Wrong angle unit

If sin 30 gives −0.988, the calculator is in radians. Switch to degrees in SETUP.

Mistake 6: -3^2 vs $(-3)^2$

$-3^2 = -9$ because the square is applied first. Type $(-3)^2$ to get 9.

Mistake 7: Division precedence

1 ÷ 2π may be read as 1 ÷ (2π) on textbook models. Type 1 ÷ (2 × π) explicitly to be sure.

Mistake 8: Forgetting SHIFT is one-shot

SHIFT applies only to the next key. If the S indicator is still lit, press SHIFT again to cancel.

Mistake 9: Clearing memory by accident

AC clears the line but not memory; CLR (SHIFT 9) can clear memory and variables.

Mistake 10: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 11: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

9 · Practice drills

Set A — Calculation drills (88)

Work each one on calculator #20, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. 25 + 5 × 6 = _____
2. 7 + 8 × 9 = _____
3. 150 + 12 × 11 = _____
4. 3 + 14 × 15 = _____
5. 48 + 6 × 7 = _____
6. 1000 + 25 × 4 = _____
7. 19 + 21 × 3 = _____
8. 64 + 8 × 8 = _____
9. (45 + 15) ÷ 6 = _____
10. (9 + 7) ÷ 2 = _____
11. (100 + 60) ÷ 8 = _____
12. (33 + 17) ÷ 5 = _____
13. (250 + 350) ÷ 12 = _____
14. (5 + 6) × (10 − 3) = _____
15. (12 + 3) × (8 − 5) = _____
16. (1.5 + 2.5) × (7 − 4) = _____

17. (20 + 30) × (15 − 5) = _____

18. (9 + 1) × (100 − 1) = _____

19. (11 + 13) × (17 − 9) = _____

20. 25^2 = _____

21. 7^2 = _____

22. 1.5^2 = _____

23. 19^2 = _____

24. 33^2 = _____

25. 0.5^2 = _____

26. 125^2 = _____

27. 64^2 = _____

28. $\sqrt{625}$ = _____

29. $\sqrt{1024}$ = _____

30. $\sqrt{9801}$ = _____

31. 3^5 = _____

32. 1.05^3 = _____

33. 5^4 = _____

34. 7^3 = _____

35. 1.5^2 = _____

36. 10^6 = _____

37. $2^{0.5}$ = _____

38. $\log 100$ = _____

39. $\log 10000$ = _____

40. $\ln 10$ = _____

41. $\ln 2$ = _____

42. $2\times10^3 \times 4$ = _____

43. $1.5\times10^4 \times 2$ = _____

44. $3\times10^5 \times 3$ = _____

45. $4.5\times10^2 \times 2$ = _____

46. $\pi \times 1.2^2$ = _____

47. $6!$ = _____

48. $0!$ = _____

49. 15×6 = , then Ans $\div 9$ = _____

50. 7.5×8 = , then Ans $\div 3$ = _____

51. 250×12 = , then Ans $\div 100$ = _____

52. $(12 + 15 + 18) \div 3$ = _____

53. $(70 + 85 + 91) \div 3$ = _____

54. $(2.5 + 3.5 + 6) \div 3$ = _____

55. $(100 + 200 + 600) \div 3$ = _____

56. $100 - 3 \times 12 + 9$ = _____

57. $18 - 2 \times 5 + 40 =$ _____

58. $7.5 - 1.5 \times 3 + 2 =$ _____

59. $64 - 8 \times 4 + 10 =$ _____

60. $300 - 25 \times 4 + 55 =$ _____

61. $9 - 3 \times 3 + 1 =$ _____

62. $45 - 5 \times 5 + 15 =$ _____

63. $1000 \div 8 \div 5 =$ _____

64. $96 \div 2 \div 3 =$ _____

65. $7200 \div 60 \div 60 =$ _____

66. $500 \div 4 \div 25 =$ _____

67. $81 \div 3 \div 3 =$ _____

68. $((10 + 2) \times 3 - 6) \div 6 =$ _____

69. $((1.5 + 2.5) \times 6 - 4) \div 5 =$ _____

70. $((20 + 30) \times 2 - 25) \div 3 =$ _____

71. $((7 + 8) \times 9 - 15) \div 12 =$ _____

72. $12 + 3 \times 4 =$ _____

73. $(12 + 8) \div 4 =$ _____

74. $(3 + 4) \times (9 - 2) =$ _____

75. $12^2 =$ _____

76. $\sqrt{144} =$ _____

77. 2^10 = _____

78. cos 0° (Deg mode) = _____

79. log 1000 = _____

80. ln 1 = _____

81. 6×10^2 × 3 = _____

82. π × 2.5^2 = _____

83. 5! = _____

84. 100 × 4 = , then Ans ÷ 2 = _____

85. (4 + 8 + 15) ÷ 3 = _____

86. 50 − 4 × 6 + 7 = _____

87. 360 ÷ 4 ÷ 9 = _____

88. ((3 + 5) × 4 − 2) ÷ 10 = _____

Set B — Key-finding drills (75)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. Which key is "▲" (Cursor ▲)? _____
2. Which key is "▼" (Cursor ▼)? _____
3. Which key is "◀" (Cursor ◀)? _____
4. Which key is "▶" (Cursor ▶)? _____
5. Which key is "⌂" (HOME)? _____
6. Which key is "OK" (Execute / =)? _____

7. Which key is "↩" (QUIT)? _____

8. Which key is "⏻" (OFF)? _____

9. Which key is "shift" (SHIFT / 2nd)? _____

10. Which key is "alpha" (ALPHA)? _____

11. Which key is "x,n,t" (The graphing variable)? _____

12. Which key is "var" (VARS)? _____

13. Which key is "toolbox" (MATH menu)? _____

14. Which key is "⌫" (DEL)? _____

15. Which key is "ex" (e^x)? _____

16. Which key is "ln" (Natural logarithm ln)? _____

17. Which key is "log" (Common logarithm $\log_{10}$)? _____

18. Which key is "i" (Imaginary unit i)? _____

19. Which key is "," (Comma / argument separator)? _____

20. Which key is "xy" (Power x^y / ^)? _____

21. Which key is "sin" (Sine)? _____

22. Which key is "cos" (Cosine)? _____

23. Which key is "tan" (Tangent)? _____

24. Which key is "π" (π)? _____

25. Which key is "√" (Square root √)? _____

26. Which key is "x^2" (Square x^2)? _____

27. Which key is “(” (Opening bracket)? _____

28. Which key is “)” (Closing bracket). Closing brackets at the end of a line may be omitted)? _____

29. Which key is “×” (Multiplication ×)? _____

30. Which key is “÷” (Division ÷)? _____

31. Which key is “+” (Addition +)? _____

32. Which key is “−” (Subtraction −)? _____

33. Which key is “.” (Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored)? _____

34. Which key is “×10x” (×10ˣ / EXP / EE)? _____

35. Which key is “Ans” (Ans)? _____

36. Which key is “EXE” (Execute / =)? _____

37. How do you reach “:”? _____

38. How do you reach “paste”? _____

39. How do you reach “clear”? _____

40. How do you reach “[”? _____

41. How do you reach “A”? _____

42. How do you reach “]”? _____

43. How do you reach “B”? _____

44. How do you reach “{”? _____

45. How do you reach “C”? _____

46. How do you reach "}"? _____

47. How do you reach "D"? _____

48. How do you reach "E"? _____

49. How do you reach " → "? _____

50. How do you reach "F"? _____

51. How do you reach "asin"? _____

52. How do you reach "G"? _____

53. How do you reach "acos"? _____

54. How do you reach "H"? _____

55. How do you reach "atan"? _____

56. How do you reach "I"? _____

57. How do you reach "="? _____

58. How do you reach "J"? _____

59. How do you reach "K"? _____

60. How do you reach "L"? _____

61. How do you reach "M"? _____

62. How do you reach "N"? _____

63. How do you reach "O"? _____

64. How do you reach "P"? _____

65. How do you reach "Q"? _____

66. How do you reach “R”? _____

67. How do you reach “S”? _____

68. How do you reach “T”? _____

69. How do you reach “U”? _____

70. How do you reach “V”? _____

71. How do you reach “W”? _____

72. How do you reach “X”? _____

73. How do you reach “Y”? _____

74. How do you reach “Z”? _____

75. How do you reach “!”? _____

Appendix A — Drill answers

Set A

1. **55** (display `55`)

   `#37` 1 *(open the calculation app from the main menu)* → `#38` 2 → `#33` 5 → `#40` + → `#33` 5 → `#35` × → `#34` 6 → `#6` OK

2. **79** (display `79`)

   `#37` 1 *(open the calculation app from the main menu)* → `#27` 7 → `#40` + → `#28` 8 → `#35` × → `#29` 9 → `#6` OK

3. **282** (display `282`)

   `#37` 1 *(open the calculation app from the main menu)* → `#37` 1 → `#33` 5 → `#42` 0 → `#40` + → `#37` 1 → `#38` 2 → `#35` × → `#37` 1 → `#37` 1 → `#6` OK

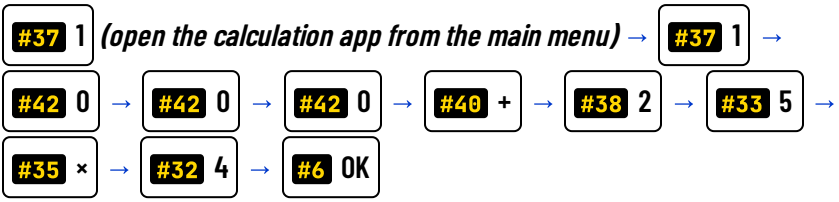
4. **213** (display `213`)

   `#37` 1 *(open the calculation app from the main menu)* → `#39` 3 → `#40` + → `#37` 1 → `#32` 4 → `#35` × → `#37` 1 → `#33` 5 → `#6` OK

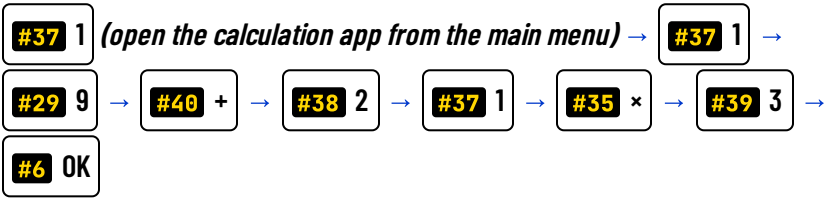
5. **90** (display `90`)

   `#37` 1 *(open the calculation app from the main menu)* → `#32` 4 → `#28` 8 → `#40` + → `#34` 6 → `#35` × → `#27` 7 → `#6` OK

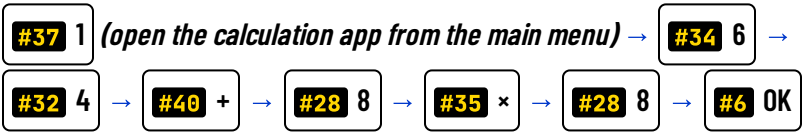
6. **1100** (display `1100`)



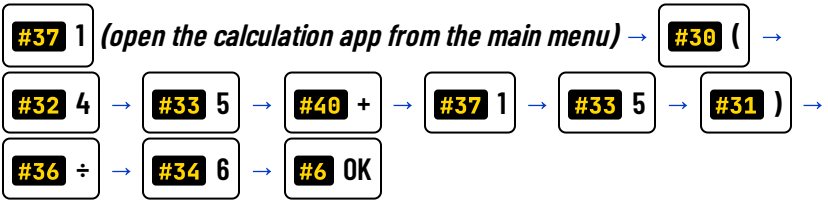
7. **82** (display `82`)



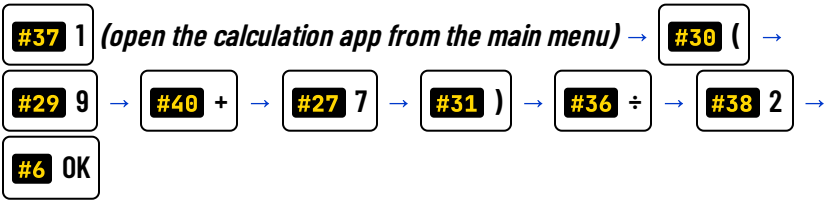
8. **128** (display `128`)



9. **10** (display `10`)



10. **8** (display `8`)



11. 20 (display 20)

#37 1 *(open the calculation app from the main menu)* → #30 (→
#37 1 → #42 0 → #42 0 → #40 + → #34 6 → #42 0 →
#31) → #36 ÷ → #28 8 → #6 OK

12. 10 (display 10)

#37 1 *(open the calculation app from the main menu)* → #30 (→
#39 3 → #39 3 → #40 + → #37 1 → #27 7 → #31) →
#36 ÷ → #33 5 → #6 OK

13. 50 (display 50)

#37 1 *(open the calculation app from the main menu)* → #30 (→
#38 2 → #33 5 → #42 0 → #40 + → #39 3 → #33 5 →
#42 0 → #31) → #36 ÷ → #37 1 → #38 2 → #6 OK

14. 77 (display 77)

#37 1 *(open the calculation app from the main menu)* → #30 (→
#33 5 → #40 + → #34 6 → #31) → #35 × → #30 (→
#37 1 → #42 0 → #41 − → #39 3 → #31) → #6 OK

15. 45 (display 45)

#37 1 *(open the calculation app from the main menu)* → #30 (→
#37 1 → #38 2 → #40 + → #39 3 → #31) → #35 × →
#30 (→ #28 8 → #41 − → #33 5 → #31) → #6 OK

16. 12 (display 12)

#37 1 *(open the calculation app from the main menu)* → #30 (→
#37 1 → #43 . → #33 5 → #40 + → #38 2 → #43 . →
#33 5 → #31) → #35 × → #30 (→ #27 7 → #41 − →
#32 4 → #31) → #6 OK

17. 500 (display 500)

#37 1 *(open the calculation app from the main menu)* → #30 (→
#38 2 → #42 0 → #40 + → #39 3 → #42 0 → #31) →
#35 × → #30 (→ #37 1 → #33 5 → #41 − → #33 5 →
#31) → #6 OK

18. 990 (display 990)

#37 1 *(open the calculation app from the main menu)* → #30 (→
#29 9 → #40 + → #37 1 → #31) → #35 × → #30 (→
#37 1 → #42 0 → #42 0 → #41 − → #37 1 → #31) →
#6 OK

19. 192 (display 192)

#37 1 *(open the calculation app from the main menu)* → #30 (→
#37 1 → #37 1 → #40 + → #37 1 → #39 3 → #31) →
#35 × → #30 (→ #37 1 → #27 7 → #41 − → #29 9 →
#31) → #6 OK

20. 625 (display `625`)

#37 1 *(open the calculation app from the main menu)* → #38 2 → #33 5 → #26 x^2 → #6 OK

21. 49 (display `49`)

#37 1 *(open the calculation app from the main menu)* → #27 7 → #26 x^2 → #6 OK

22. 2.25 (display `2.25`)

#37 1 *(open the calculation app from the main menu)* → #37 1 → #43 . → #33 5 → #26 x^2 → #6 OK

23. 361 (display `361`)

#37 1 *(open the calculation app from the main menu)* → #37 1 → #29 9 → #26 x^2 → #6 OK

24. 1089 (display `1089`)

#37 1 *(open the calculation app from the main menu)* → #39 3 → #39 3 → #26 x^2 → #6 OK

25. 0.25 (display `0.25`)

#37 1 *(open the calculation app from the main menu)* → #42 0 → #43 . → #33 5 → #26 x^2 → #6 OK

26. 15625 (display `15625`)

#37 1 *(open the calculation app from the main menu)* → #37 1 → #38 2 → #33 5 → #26 x^2 → #6 OK

27. 4096 (display `4096`)

#37 1 *(open the calculation app from the main menu)* → #34 6 → #32 4 → #26 x^2 → #6 OK

28. 25 (display `25`)

#37 1 *(open the calculation app from the main menu)* → #25 √ → #34 6 → #38 2 → #33 5 → #6 OK

29. 32 (display `32`)

#37 1 *(open the calculation app from the main menu)* → #25 √ → #37 1 → #42 0 → #38 2 → #32 4 → #6 OK

30. 99 (display `99`)

#37 1 *(open the calculation app from the main menu)* → #25 √ → #29 9 → #28 8 → #42 0 → #37 1 → #6 OK

31. 243 (display `243`)

#37 1 *(open the calculation app from the main menu)* → #39 3 → #20 x^y → #33 5 → #6 OK

32. 1.157625 (display `1.157625`)

#37 1 *(open the calculation app from the main menu)* → #37 1 → #43 . → #42 0 → #33 5 → #20 x^y → #39 3 → #6 OK

33. 625 (display `625`)

`#37` 1 *(open the calculation app from the main menu)* → `#33` 5 → `#20` x^y → `#32` 4 → `#6` OK

34. 343 (display `343`)

`#37` 1 *(open the calculation app from the main menu)* → `#27` 7 → `#20` x^y → `#39` 3 → `#6` OK

35. 2.25 (display `2.25`)

`#37` 1 *(open the calculation app from the main menu)* → `#37` 1 → `#43` . → `#33` 5 → `#20` x^y → `#38` 2 → `#6` OK

36. 1000000.000 (display `1000000`)

`#37` 1 *(open the calculation app from the main menu)* → `#37` 1 → `#42` 0 → `#20` x^y → `#34` 6 → `#6` OK

37. 1.41421356 (display `1.414213562`)

`#37` 1 *(open the calculation app from the main menu)* → `#38` 2 → `#20` x^y → `#42` 0 → `#43` . → `#33` 5 → `#6` OK

38. 2 (display `2`)

`#37` 1 *(open the calculation app from the main menu)* → `#17` log → `#37` 1 → `#42` 0 → `#42` 0 → `#31`) → `#6` OK

39. 4 (display `4`)

#37 1 *(open the calculation app from the main menu)* → #17 log → #37 1 → #42 0 → #42 0 → #42 0 → #42 0 → #31) → #6 OK

40. 2.30258509 (display `2.302585093`)

#37 1 *(open the calculation app from the main menu)* → #16 ln → #37 1 → #42 0 → #31) → #6 OK

41. 0.69314718 (display `0.6931471806`)

#37 1 *(open the calculation app from the main menu)* → #16 ln → #38 2 → #31) → #6 OK

42. 8000 (display `8000`)

#37 1 *(open the calculation app from the main menu)* → #38 2 → #44 ×10^x → #39 3 → #35 × → #32 4 → #6 OK

43. 30000 (display `30000`)

#37 1 *(open the calculation app from the main menu)* → #37 1 → #43 . → #33 5 → #44 ×10^x → #32 4 → #35 × → #38 2 → #6 OK

44. 900000 (display `900000`)

#37 1 *(open the calculation app from the main menu)* → #39 3 → #44 ×10^x → #33 5 → #35 × → #39 3 → #6 OK

45. 900 (display `900`)

#37 1 *(open the calculation app from the main menu)* → #32 4 → #43 . → #33 5 → #44 ×10^x → #38 2 → #35 × → #38 2 → #6 OK

46. 4.52389342 (display `4.523893421`)

#37 1 *(open the calculation app from the main menu)* → #24 π → #35 × → #37 1 → #43 . → #38 2 → #26 x^2 → #6 OK

47. 720 (display `720`)

#37 1 *(open the calculation app from the main menu)* → #34 6 → #10 alpha → #43 ! → #6 OK

48. 1 (display `1`)

#37 1 *(open the calculation app from the main menu)* → #42 0 → #10 alpha → #43 ! → #6 OK

49. 10 (display `10`)

#37 1 *(open the calculation app from the main menu)* → #37 1 → #33 5 → #35 × → #34 6 → #6 OK → #36 ÷ → #29 9 → #6 OK

50. 20 (display `20`)

#37 1 *(open the calculation app from the main menu)* → #27 7 → #43 . → #33 5 → #35 × → #28 8 → #6 OK → #36 ÷ → #39 3 → #6 OK

51. 30 (display 30)

#37 1 *(open the calculation app from the main menu)* → #38 2 → #33 5 → #42 0 → #35 × → #37 1 → #38 2 → #6 OK → #36 ÷ → #37 1 → #42 0 → #42 0 → #6 OK

52. 15 (display 15)

#37 1 *(open the calculation app from the main menu)* → #30 (→ #37 1 → #38 2 → #40 + → #37 1 → #33 5 → #40 + → #37 1 → #28 8 → #31) → #36 ÷ → #39 3 → #6 OK

53. 82 (display 82)

#37 1 *(open the calculation app from the main menu)* → #30 (→ #27 7 → #42 0 → #40 + → #28 8 → #33 5 → #40 + → #29 9 → #37 1 → #31) → #36 ÷ → #39 3 → #6 OK

54. 4 (display 4)

#37 1 *(open the calculation app from the main menu)* → #30 (→ #38 2 → #43 . → #33 5 → #40 + → #39 3 → #43 . → #33 5 → #40 + → #34 6 → #31) → #36 ÷ → #39 3 → #6 OK

55. 300 (display 300)

#37 1 *(open the calculation app from the main menu)* → #30 (→
#37 1 → #42 0 → #42 0 → #40 + → #38 2 → #42 0 →
#42 0 → #40 + → #34 6 → #42 0 → #42 0 → #31) →
#36 ÷ → #39 3 → #6 OK

56. 73 (display 73)

#37 1 *(open the calculation app from the main menu)* → #37 1 →
#42 0 → #42 0 → #41 − → #39 3 → #35 × → #37 1 →
#38 2 → #40 + → #29 9 → #6 OK

57. 48 (display 48)

#37 1 *(open the calculation app from the main menu)* → #37 1 →
#28 8 → #41 − → #38 2 → #35 × → #33 5 → #40 + →
#32 4 → #42 0 → #6 OK

58. 5 (display 5)

#37 1 *(open the calculation app from the main menu)* → #27 7 →
#43 . → #33 5 → #41 − → #37 1 → #43 . → #33 5 →
#35 × → #39 3 → #40 + → #38 2 → #6 OK

59. 42 (display 42)

#37 1 *(open the calculation app from the main menu)* → #34 6 →
#32 4 → #41 − → #28 8 → #35 × → #32 4 → #40 + →
#37 1 → #42 0 → #6 OK

60. 255 (display `255`)

#37 1 *(open the calculation app from the main menu)* → #39 3 → #42 0 → #42 0 → #41 − → #38 2 → #33 5 → #35 × → #32 4 → #40 + → #33 5 → #33 5 → #6 OK

61. 1 (display `1`)

#37 1 *(open the calculation app from the main menu)* → #29 9 → #41 − → #39 3 → #35 × → #39 3 → #40 + → #37 1 → #6 OK

62. 35 (display `35`)

#37 1 *(open the calculation app from the main menu)* → #32 4 → #33 5 → #41 − → #33 5 → #35 × → #33 5 → #40 + → #37 1 → #33 5 → #6 OK

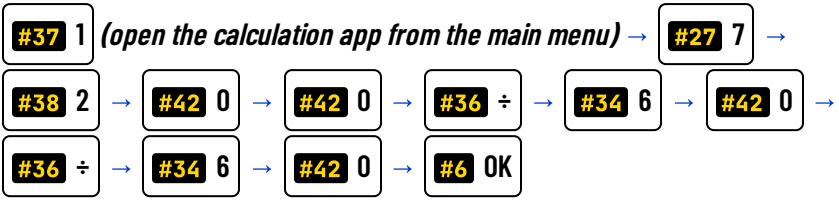
63. 25 (display `25`)

#37 1 *(open the calculation app from the main menu)* → #37 1 → #42 0 → #42 0 → #42 0 → #36 ÷ → #28 8 → #36 ÷ → #33 5 → #6 OK

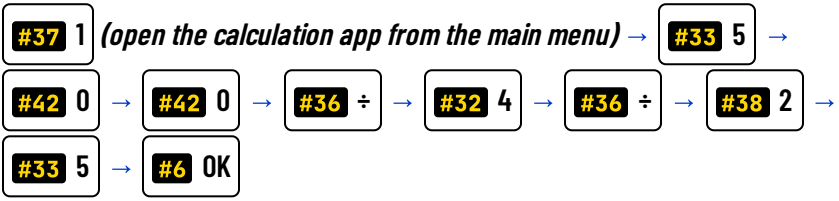
64. 16 (display `16`)

#37 1 *(open the calculation app from the main menu)* → #29 9 → #34 6 → #36 ÷ → #38 2 → #36 ÷ → #39 3 → #6 OK

65. 2 (display `2`)



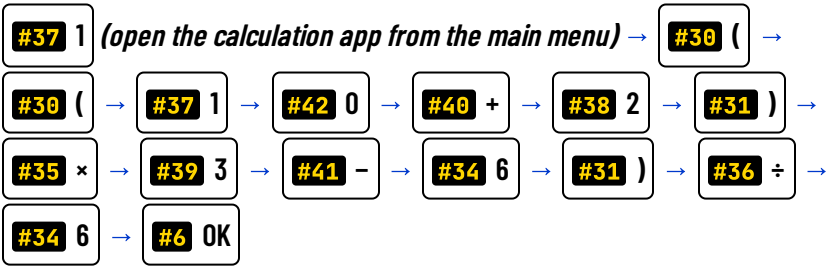
66. 5 (display `5`)



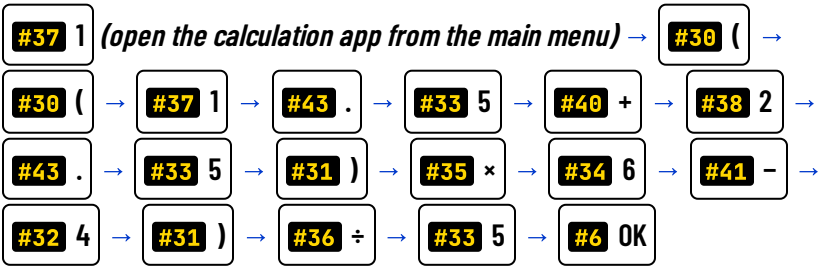
67. 9 (display `9`)



68. 5 (display `5`)



69. 4 (display `4`)



70. 25 (display 25)

#37 1 *(open the calculation app from the main menu)* → #30 (→
#30 (→ #38 2 → #42 0 → #40 + → #39 3 → #42 0 →
#31) → #35 × → #38 2 → #41 − → #38 2 → #33 5 →
#31) → #36 ÷ → #39 3 → #6 OK

71. 10 (display 10)

#37 1 *(open the calculation app from the main menu)* → #30 (→
#30 (→ #27 7 → #40 + → #28 8 → #31) → #35 × →
#29 9 → #41 − → #37 1 → #33 5 → #31) → #36 ÷ →
#37 1 → #38 2 → #6 OK

72. 24 (display 24)

#37 1 *(open the calculation app from the main menu)* → #37 1 →
#38 2 → #40 + → #39 3 → #35 × → #32 4 → #6 OK

73. 5 (display 5)

#37 1 *(open the calculation app from the main menu)* → #30 (→
#37 1 → #38 2 → #40 + → #28 8 → #31) → #36 ÷ →
#32 4 → #6 OK

74. 49 (display 49)

#37 1 *(open the calculation app from the main menu)* → #30 (→
#39 3 → #40 + → #32 4 → #31) → #35 × → #30 (→
#29 9 → #41 − → #38 2 → #31) → #6 OK

75. 144 (display `144`)

#37 1 *(open the calculation app from the main menu)* → #37 1 → #38 2 → #26 x^2 → #6 OK

76. 12 (display `12`)

#37 1 *(open the calculation app from the main menu)* → #25 √ → #37 1 → #32 4 → #32 4 → #6 OK

77. 1024 (display `1024`)

#37 1 *(open the calculation app from the main menu)* → #38 2 → #20 x^y → #37 1 → #42 0 → #6 OK

78. 1 (display `1`)

#37 1 *(open the calculation app from the main menu)* → #22 cos → #42 0 → #31) → #6 OK

79. 3 (display `3`)

#37 1 *(open the calculation app from the main menu)* → #17 log → #37 1 → #42 0 → #42 0 → #42 0 → #31) → #6 OK

80. 0 (display `0`)

#37 1 *(open the calculation app from the main menu)* → #16 ln → #37 1 → #31) → #6 OK

81. 1800 (display `1800`)

`#37` 1 *(open the calculation app from the main menu)* → `#34` 6 → `#44` ×10^x → `#38` 2 → `#35` × → `#39` 3 → `#6` OK

82. 19.63495408 (display `19.63495408`)

`#37` 1 *(open the calculation app from the main menu)* → `#24` π → `#35` × → `#38` 2 → `#43` . → `#33` 5 → `#26` x^2 → `#6` OK

83. 120 (display `120`)

`#37` 1 *(open the calculation app from the main menu)* → `#33` 5 → `#10` alpha → `#43` ! → `#6` OK

84. 200 (display `200`)

`#37` 1 *(open the calculation app from the main menu)* → `#37` 1 → `#42` 0 → `#42` 0 → `#35` × → `#32` 4 → `#6` OK → `#36` ÷ → `#38` 2 → `#6` OK

85. 9 (display `9`)

`#37` 1 *(open the calculation app from the main menu)* → `#30` (→ `#32` 4 → `#40` + → `#28` 8 → `#40` + → `#37` 1 → `#33` 5 → `#31`) → `#36` ÷ → `#39` 3 → `#6` OK

86. 33 (display `33`)

`#37` 1 *(open the calculation app from the main menu)* → `#33` 5 → `#42` 0 → `#41` − → `#32` 4 → `#35` × → `#34` 6 → `#40` + → `#27` 7 → `#6` OK

87. 10 (display 10)

#37 1 *(open the calculation app from the main menu)* → #39 3 → #34 6 → #42 0 → #36 ÷ → #32 4 → #36 ÷ → #29 9 → #6 OK

88. 3 (display 3)

#37 1 *(open the calculation app from the main menu)* → #30 (→ #30 (→ #39 3 → #40 + → #33 5 → #31) → #35 × → #32 4 → #41 − → #38 2 → #31) → #36 ÷ → #37 1 → #42 0 → #6 OK

Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14
15. #15
16. #16
17. #17
18. #18
19. #19
20. #20
21. #21
22. #22
23. #23
24. #24
25. #25
26. #26
27. #30
28. #31
29. #35
30. #36
31. #40
32. #41
33. #43
34. #44
35. #45
36. #46
37. #10 alpha then #11 x,n,t
38. #9 shift then #13 toolbox
39. #9 shift then #14 ⌫
40. #9 shift then #15 ex
41. #10 alpha then #15 ex
42. #9 shift then #16 ln

43. #10 alpha then #16 ln
44. #9 shift then #17 log
45. #10 alpha then #17 log
46. #9 shift then #18 i
47. #10 alpha then #18 i
48. #10 alpha then #19 ,
49. #9 shift then #20 xy
50. #10 alpha then #20 xy
51. #9 shift then #21 sin
52. #10 alpha then #21 sin
53. #9 shift then #22 cos
54. #10 alpha then #22 cos
55. #9 shift then #23 tan
56. #10 alpha then #23 tan
57. #9 shift then #24 π
58. #10 alpha then #24 π
59. #10 alpha then #25 √
60. #10 alpha then #26 x^2
61. #10 alpha then #27 7
62. #10 alpha then #28 8
63. #10 alpha then #29 9
64. #10 alpha then #30 (
65. #10 alpha then #31)
66. #10 alpha then #32 4
67. #10 alpha then #33 5
68. #10 alpha then #34 6
69. #10 alpha then #35 ×
70. #10 alpha then #36 ÷
71. #10 alpha then #37 1
72. #10 alpha then #38 2
73. #10 alpha then #39 3
74. #10 alpha then #40 +
75. #10 alpha then #43 .

Appendix B — Function finder (A–Z)

Function	Label	Keys
×10^x / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10^x 23	×10x	#44
Addition +	+	#40
ALPHA — the next key types its letter/variable (red/green label)	alpha	#10
Ans — recall the last answer into the current expression	Ans	#45
CLEAR — clear the entry line / screen	clear	#9 → #14
Closing bracket). Closing brackets at the end of a line may be omitted	)	#31
Colon : (multi-statement separator)	:	#10 → #11
Comma / argument separator (between function arguments, list items or coordinates)	,	#19
Common logarithm $\log_{10}$	log	#17
Cosine (current angle unit)	cos	#22
Cursor ▲ — up / previous history line / previous item	▲	#1
Cursor ▶ — move right in the line	▶	#4
Cursor ▼ — down / next history line / next item	▼	#2
Cursor ◀ — move left in the line	◀	#3
Decimal point. Pressing it first gives "0."; a second point in the same number is ignored	.	#43
DEL — delete the character at the cursor	⌫	#14
Digit 0	0	#42
Digit 1	1	#37
Digit 2	2	#38
Digit 3	3	#39
Digit 4	4	#32
Digit 5	5	#33

Digit 6	6	#34
Digit 7	7	#27
Digit 8	8	#28
Digit 9	9	#29
Division ÷	÷	#36
ENTRY — recall a previous entry	paste	#9 → #13
Equals — complete the calculation	=	#9 → #24
e^x — exponential	ex	#15
Execute / = — evaluate the line and show the result (EXE / ENTER / =)	OK	#6
Execute / = — evaluate the line and show the result (EXE / ENTER / =)	EXE	#46
Factorial n! (also on non-integers via the gamma function where supported)	!	#10 → #43
HOME — return to the main calculation screen	⌂	#5
Imaginary unit i (complex numbers)	i	#18
Inverse cosine $\cos^{-1}$	acos	#9 → #22
Inverse sine $\sin^{-1}$	asin	#9 → #21
Inverse tangent $\tan^{-1}$	atan	#9 → #23
MATH menu — extra math functions	toolbox	#13
Multiplication ×	×	#35
Natural logarithm ln	ln	#16
OFF — power off (state is lost; press ON to restart)	⏻	#8
Opening bracket (	(	#30
Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)	xy	#20
QUIT — leave the current screen / app	↩	#7
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	shift	#9

Sine (uses the current angle unit)	sin	#21
Square root √	√	#25
Square x^2	x^2	#26
STO▶ (→) — store into a variable	→	#9 → #20
Subtraction –	–	#41
Tangent (current angle unit)	tan	#23
The graphing variable (X / x,θ,t,n) for Y= functions, tables and solving	x,n,t	#11
Types "["	[	#9 → #15
Types "]"	]	#9 → #16
Types "{"	{	#9 → #17
Types "}"	}	#9 → #18
Variable A (store with STO/ →, recall by name)	A	#10 → #15
Variable B (store with STO/ →, recall by name)	B	#10 → #16
Variable C (store with STO/ →, recall by name)	C	#10 → #17
Variable D (store with STO/ →, recall by name)	D	#10 → #18
Variable E (store with STO/ →, recall by name)	E	#10 → #19
Variable F (store with STO/ →, recall by name)	F	#10 → #20
Variable G (store with STO/ →, recall by name)	G	#10 → #21
Variable H (store with STO/ →, recall by name)	H	#10 → #22

Variable I (store with STO/ →, recall by name)	**I**	**#10 → #23**
Variable J (store with STO/ →, recall by name)	**J**	**#10 → #24**
Variable K (store with STO/ →, recall by name)	**K**	**#10 → #25**
Variable L (store with STO/ →, recall by name)	**L**	**#10 → #26**
Variable M (store with STO/ →, recall by name)	**M**	**#10 → #27**
Variable N (store with STO/ →, recall by name)	**N**	**#10 → #28**
Variable O (store with STO/ →, recall by name)	**O**	**#10 → #29**
Variable P (store with STO/ →, recall by name)	**P**	**#10 → #30**
Variable Q (store with STO/ →, recall by name)	**Q**	**#10 → #31**
Variable R (store with STO/ →, recall by name)	**R**	**#10 → #32**
Variable S (store with STO/ →, recall by name)	**S**	**#10 → #33**
Variable T (store with STO/ →, recall by name)	**T**	**#10 → #34**
Variable U (store with STO/ →, recall by name)	**U**	**#10 → #35**
Variable V (store with STO/ →, recall by name)	**V**	**#10 → #36**
Variable W (store with STO/ →, recall by name)	**W**	**#10 → #37**
Variable X (store with STO/ →, recall by name)	**X**	**#10 → #38**

Variable Y (store with STO/ →, recall by name)	Y	#10 → #39
Variable Z (store with STO/ →, recall by name)	Z	#10 → #40
VARS — variables menu	var	#12
π (3.14159265…) as an exact constant	π	#24

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

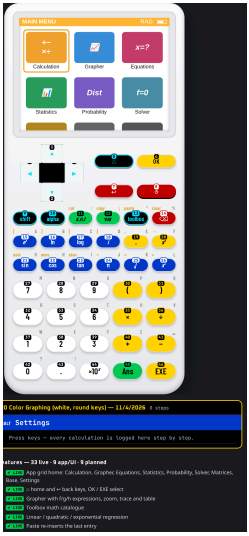
Round-key colour graphing model. Starts on the app grid (Calculation, Grapher, Equations, Statistics, Probability, Solver, Matrices, Base, Settings); ⌂ returns to it, ↩ goes back, OK/EXE select. Grapher has f/g/h… expressions, graph with zoom & trace, and a table. Toolbox is the math catalogue. Python, sequences and regression model choice beyond linear/quadratic/exponential are not simulated; cut/copy are no-ops (paste re-inserts the last entry).

Planned (not yet working)

- Python scripts
- Sequences app
- More regression models
- Cut / copy clipboard
- Graph image export (PNG)
- Shareable function list link
- Dark screen theme
- Speech read-out of values
- Exam mode

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 ▲	#2 ▼	#3 ◀
#4 ▶	#5 ⌂	#6 OK
#7 ↩	#8 ⏻	#9 shift
#10 alpha	#11 x,n,t ⇧ cut	#12 var ⇧ copy
#13 toolbox ⇧ paste	#14 ⌫ ⇧ clear	#15 ex ⇧ [
#16 ln ⇧]	#17 log ⇧ {	#18 i ⇧ }
#19 , ⇧ _	#20 xy ⇧ →	#21 sin ⇧ asin
#22 cos ⇧ acos	#23 tan ⇧ atan	#24 π ⇧ =
#25 √ ⇧ <	#26 x^2 ⇧ >	#27 7
#28 8	#29 9	#30 (
#31)	#32 4	#33 5
#34 6	#35 ×	#36 ÷
#37 1	#38 2	#39 3
#40 +	#41 –	#42 0
#43 .	#44 ×10x	#45 Ans
#46 EXE		

U-45 CALCULATORS · Calculator #20 · cut out and keep